CSEC PHYSICS

May/Jun 2023 – Paper 2

Solutions

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SECTION A

Answer ALL questions.

1. Figure 1 shows the setup for an experiment to determine the specific heat capacity of 300g of a liquid using an electrical heater. The heater has a power rating of 105 W.

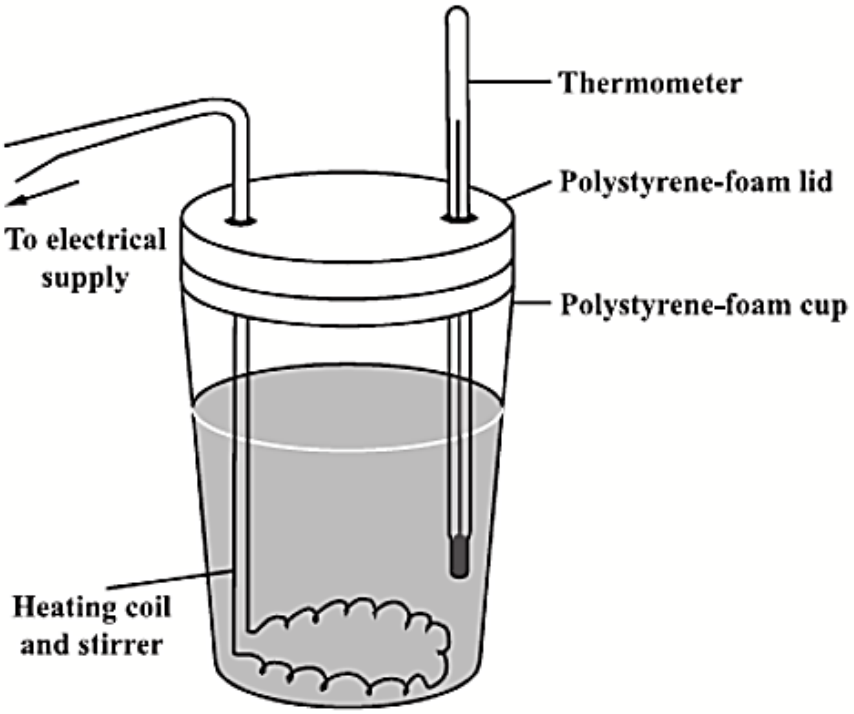

Figure 1. Experiment to determine the specific heat capacity of a liquid.

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Temperature and time data for the liquid in the polystyrene-foam cup are shown in Table 1

TABLE 1: TEMPERATURE, θ, AND CORRESPONDING TIME, t

Temperature $\theta/^0$C	Time t/seconds
25.3	0
30.5	60
34.9	120
41.2	180
46.4	240
51.5	300
56.8	360

(a) Define the term 'specific heat capacity' (3 marks)

The specific heat capacity, c of a substance is the heat energy required to raise the temperature of 1 kg of the substance by 1 Kelvin.

(b) Assuming that all the electrical energy produced by the heater is transferred to the liquid, and that:

P is the electrical power of the heater

t is the time duration of the experiment

m is the mass of the liquid

c is the specific heat capacity

$\Delta\theta$ is the increase in the temperature of the liquid

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Deduce the formula which represents the heat energy transferred from the heater to the liquid. (1 mark)

$$P = \frac{\Delta E}{t}$$

$$\Delta E = Pt$$

$$Therefore, Pt = mc\Delta\theta$$

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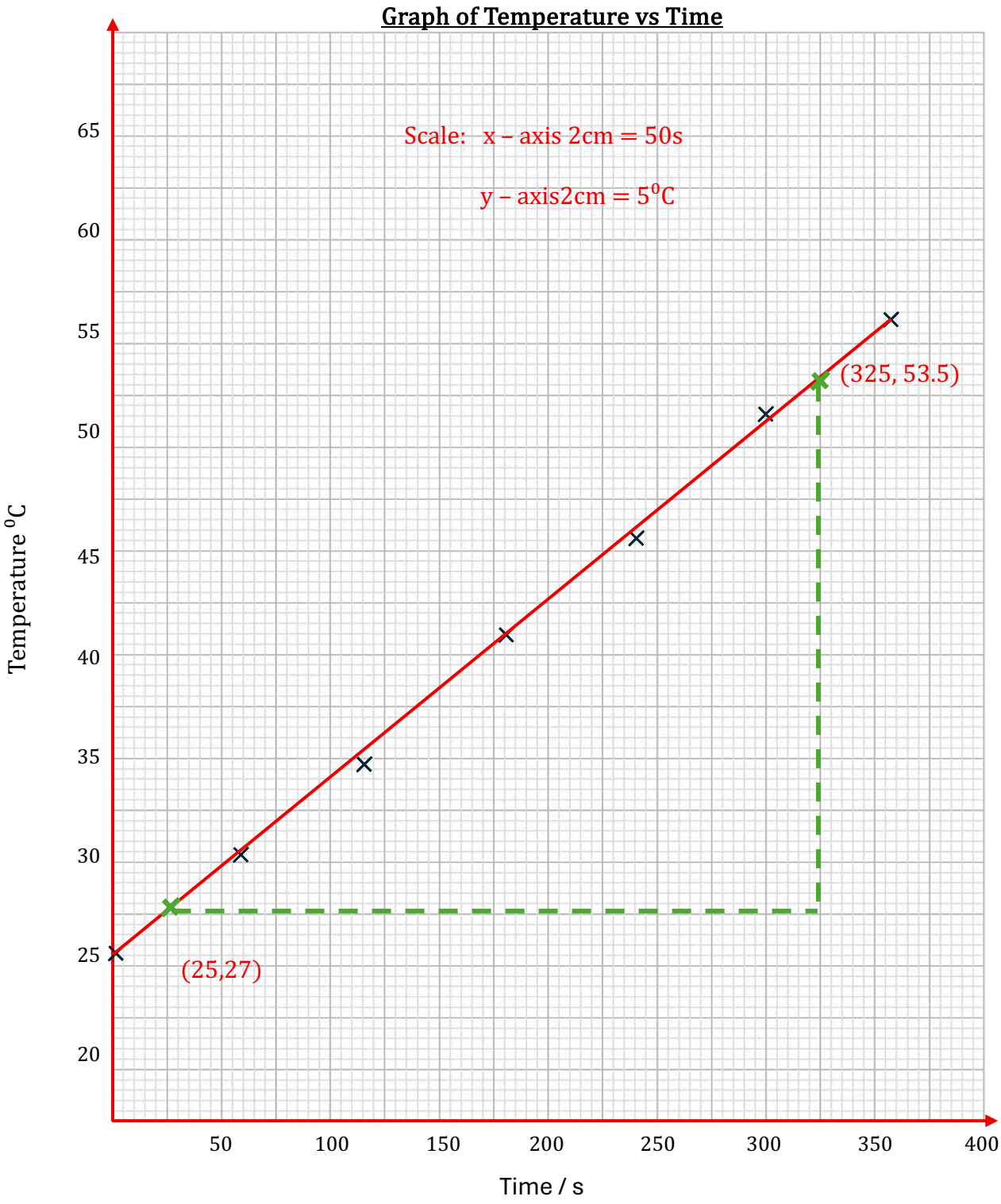
(c) Using the grid provided, plot a graph of temperature versus time. (8 marks)



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(d) Calculate the gradient, G, of the graph. (5 marks)

$$Gradient = \frac{y_2 - y_1}{x_2 - x_1}$$

Using points (25, 27) and (325, 53.5) from Graph

$$Gradient = \frac{53.5-27}{325-25}$$

$$= 0.088^0 C/s$$

(e) (i) Given that G=P/mc, calculate the specific heat capacity, of the liquid. Express your answer in S.I. units, to one significant figure. (6 marks)

P = 105W

M = 300g

= 0.3kg

c =Required to find

G= 0.088^0C/s

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Hence

$$G = \frac{P}{mc}$$

$$mc = \frac{P}{G}$$

$$c = \frac{P}{GM}$$

$$c = \frac{105}{(0.088)(0.3)}$$

$$c = 3977 Jkg^{-1}K^{-1}$$

$$c = 4000\ Jkg^{-1}K^{-1}\ to\ 1\ sf$$

(ii) Table 2 shows the specific heat capacities for three commonly used liquids.

TABLE 2: LIQUID AND CORRESPONDING SPECIFIC HEAT CAPACITIES

Liquid	Specific Heat Capacity(J/kg/^{0}C)
Cooking oil	1700
Milk	3900
Paraffin	2100

Using the answer calculated in (e)(i) and the information in the Table 2, state the liquid that was most likely used in the experiment.

Liquid: Milk (1 mark)

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(f) State ONE precaution that the class should take in the performing experiment.

(1 mark)

Ensure that the bulb of the thermometer is suspended in the liquid and not touching the heating element away from the side of the cup.

2. Figure 3 shows a wind turbine used in some Caribbean countries as an alternative energy source.

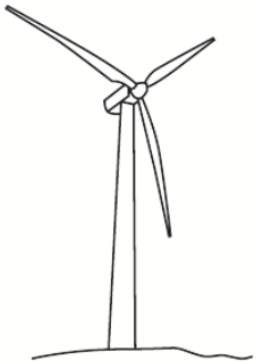

Figure 3. Diagram of a wind turbine

(a) Describe how a wind turbine produces electrical energy (3 marks)

Wind energy(kinetic energy) turns propeller blades (converted to mechanical energy) which turns a turbine (generator)which creates electrical energy

(b) State TWO advantages and TWO disadvantages of using wind turbines in some Caribbean countries (4 marks)

Advantages:

- Renewable Energy
- No pollution

Disadvantages:

- Unreliable source of energy (depends on steady wind supply which is very variable in the Caribbean)
- Possibility of Hurricanes which destroy wind turbines.

(c) If the mass of the air that passes through the circular area swept out by the turbine blades every second in 8.5kg and its kinetic energy during this time is 2500J, calculate

(i) The speed of the air (3 marks)

$$E_k = \frac{1}{2}mv^2$$

$$\frac{2E_k}{m} = v^2$$

$$v = \sqrt{\frac{2E_k}{m}}$$

$$v = \sqrt{\frac{2(2500)}{8.5}}$$

$$v = 24.25ms^{-1}$$

(ii) The input power of the air (2 marks)

$$P = \frac{\Delta E}{t}$$

$$P = \frac{2500}{1}$$

$$= 2500W$$

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(iii) The efficiency of the wind turbine, given that the output power of the generator is 1350 W. (3 marks)

$$Efficiency = \frac{Useful\ Energy\ Output}{Total\ Energy\ Input} \times 100$$

$$= \frac{1350}{2500} \times 100$$

$$= 54\%$$

Total 15 marks

3. (a) State TWO properties of electromagnetic waves (2 marks)

- Can travel in a vacuum
- Can be reflected/refracted/diffracted
- Travels at the speed of light
- Can be polarized

(b) Figure 4 shows a cup and a string telephone.

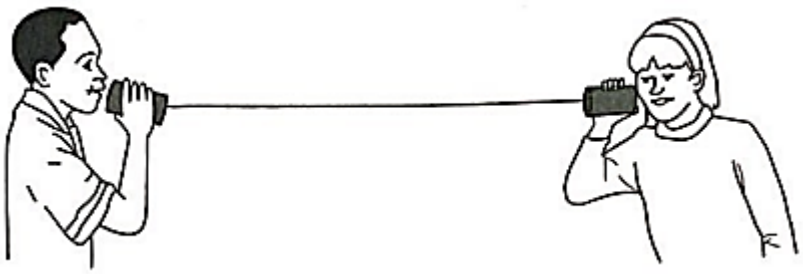

Figure 4. Diagram of a cup and string telephone

(i) Describe how the cup and string works.

- When person A speaks, the air molecules vibrate into the bottom of the cup.

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- This is transferred to the string and vibrates along the length of the string to the next cup.
- This causes the receiving cup to vibrate along with the air in the cup.
- These vibrations are heard by Person B as sound.

(ii) Calculate the wavelength of the sound produced by the cup and string telephone, given that the frequency of sound in the air is 11 000Hz and the speed of the sound in air is 330 m/s. (3 marks)

f= 11000Hz

v=330 m/s

$$V = f\lambda$$

$$\boldsymbol{\lambda = \frac{v}{f}}$$

$$= \frac{330}{11000}$$

$$= 0.03\text{m}$$

(c) Figure 5 shows a coin, at position O, in an empty container. An observer is unable to see the coin based on the position of his eye.

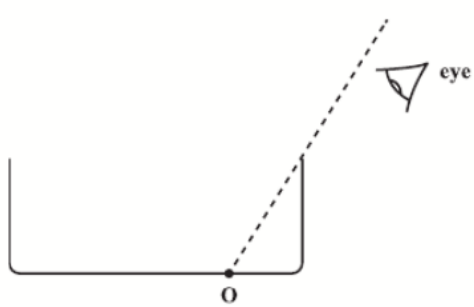

Figure 5. A coin in an empty container

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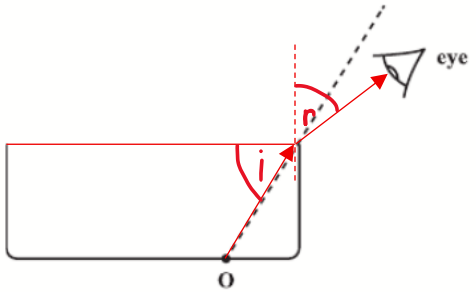
(i) Describe how the observer can see the coin when the container is filled with water (2 marks)



The light ray from the coin will now refract away from the normal since its going from a more dense to a less dense medium, which means that the angle of refraction is greater than the angle of incidence and the ray would now be seen by the eye.

(ii) The refractive index of water is defined as the ratio of the speed of the light in the air to the speed of light in water.

Calculate the speed of light in water, given that the speed of light in air is $3 \times 10^8\ m/s$ and the refractive index of water is 1.3. (2 marks)

$$_a n_w = \frac{C_a}{C_w}$$

$$C_w = \frac{C_a}{_a n_w}$$

$$= \frac{3 \times 10^8}{1.3}$$

$$= 2.31 \times 10^8 m/s$$

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(iii) Calculate the critical angle of water. (3 marks)

$$_{a}n_{w} = \frac{1}{\sin c}$$

$$\sin c = \frac{1}{_{a}n_{w}}$$

$$= \sin^{-1}\left(\frac{1}{1.3}\right)$$

$$= 50.3^{0}\ to\ 1\ dp$$

Total 15 marks

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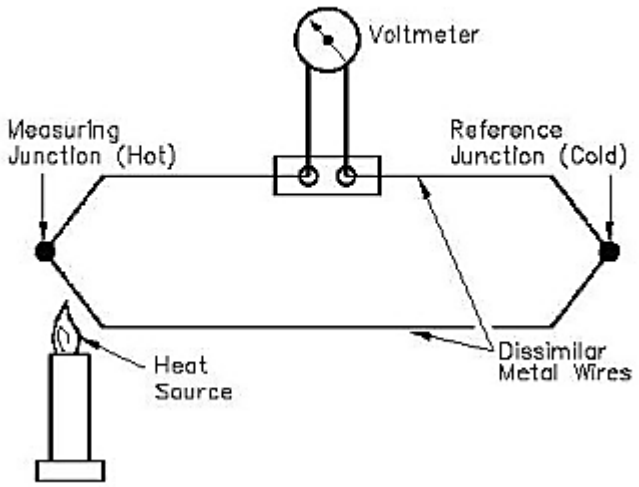
SECTION B

Answer ALL questions.

4. (a) (i) Draw a labelled diagram of the structure of a thermocouple thermometer.

(3 marks)



(ii) The thermocouple thermometer is used to measure temperature of the blue part of the flame of a small candle. State THREE reasons why a thermocouple thermometer is suitable for this application.

- Responds quickly to rapidly changing temperatures
- Can be used at very high temperatures
- Is very sensitive (increased accuracy)
- Very small and can be placed exactly in the blue part of the flame

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(3 marks)

(b) Figure 6 shows a fixed mass of gas which is trapped in a cylinder by a piston.

The gas is at atmospheric pressure $1.0 \times 10^5\ Pa$. The volume of the trapped gas is $1.9 \times 10^{-4} m^3$ and the initial temperature is 27^0C. An electrical heater is used to increase the temperature of the trapped gas to a final temperature of 627 ^{0}C.

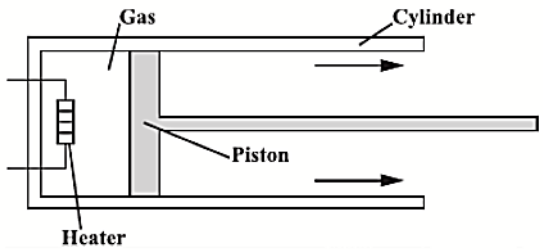

Figure 6. Diagram of a fixed mass of gas trapped in a cylinder by a piston

(i) Use the kinetic theory to explain how the gas exerts pressure on the piston.

(3 marks)

Based on the Kinetic Theory of matter, an increase in temperature results in an increase in the Kinetic energy of its particles. An increase in temperature results in the increased kinetic energy of the gas particles in the cylinder, thus increasing the frequency of collisions with the surface area of the piston exposed to the gas. Thus, this change in momentum changes the force per unit area of the gas acting on the piston and hence the pressure exerted on the piston by the heated gas increases.

(ii) Calculate the final volume of the gas, given that the final pressure is $1.8 \times 10^5 Pa$.

$P_1 = 1.0 \times 10^5 Pa$ $\quad$ $P_2 = 1.8 \times 10^5 Pa$

$V_1 = 1.9 \times 10^{-4} m^3$ $\quad$ $V_2 = unknown$

$T_1 = (27 + 273)K$ $\quad$ $T_2 = (627 + 273)K$

Gas Law: $\frac{P_2 V_2}{T_2} = \frac{P_1 V_1}{T_1}$

$$\frac{(1.8 \times 10^5)V_2}{900} = \frac{(1.0 \times 10^5)(1.9 \times 10^{-4})}{300}$$

$$V_2 = 3.17 \times 10^{-4} m^3$$

(6 marks)

Total 15 marks

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5. (a) (i) State the name of the logic gate represented in Figure 7 (1 mark)

Figure 7. Logic gate

NAND

(ii) State the name of the logic gate using the truth table shown in the table below. (1 mark)

Input	Output
0	1
1	0

NOT

(iii) Figure 8 is a digital circuit consisting of a combination of basic logic gates.

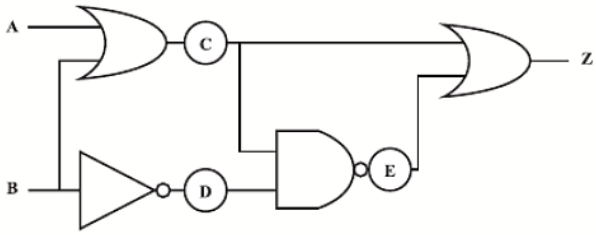

Figure 8. Digital circuit of basic logic gates

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Complete the truth table below for the circuit in Figure 8.

INPUTS		C	D	E	OUTPUT
A	B				Z
0	0	0	1	1	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	1	0	1	1

(6 marks)

(b) Figure 9 shows an arrangement for making an electromagnet.

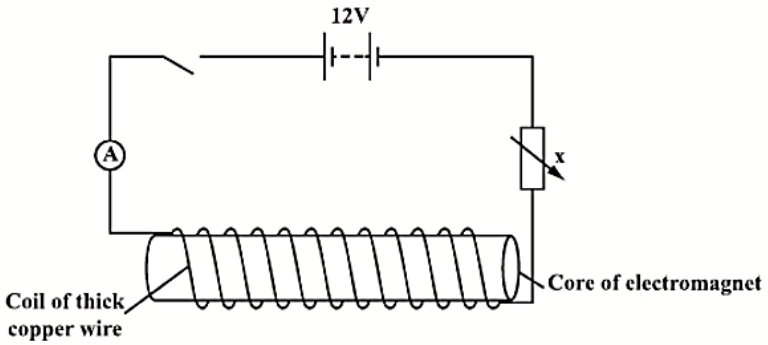

Figure 9. An electromagnet

(i) State a material which is suitable for the core of an electromagnet.

Soft Iron (1 marks)

(ii) List TWO factors which would cause an increase in the strength of an electromagnet. (2 marks)

- More coils = More turns
- Stronger battery = higher voltage

(iii) State how component X can be used to increase the strength of an electromagnet. (1 mark)

X is a Rheostat or a variable resistor. Reducing resistance increases current which increases the strength of the electromagnet.

(iv) When the switch is closed, the reading on the ammeter is 1.5A. Calculate the resistance of the circuit. (3 marks)

$$V = IR$$

$$R = \frac{V}{I}$$

$$= \frac{12}{1.5}$$

$$= 8\ \Omega$$

Total 15 marks

6. Figure 10 shows a vacuum tube with a radioactive source which emits α- particles, β particles and γ rays through an electric field.

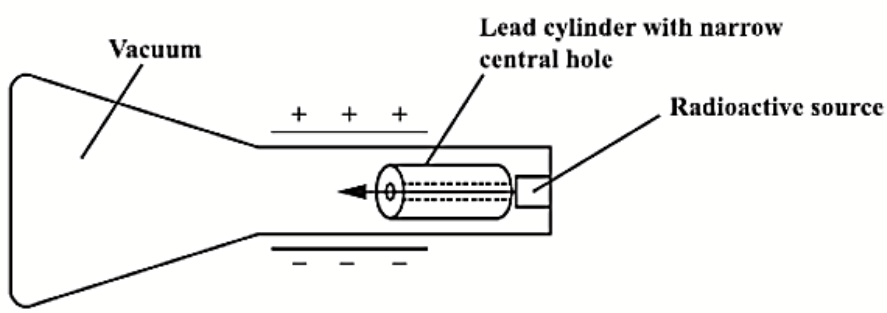

Figure 10. Vacuum tube with a radioactive source emitting radiation which passes through an electric field.

(a) Describe the paths of the α- particles, β particles and γ rays as they pass through the electric field. State a reason for EACH answer.

(i) α- particles (2 marks)

These are deflected toward the negative plate since the α- particles are positively charged. (slower deflection since they are heavier than β particles)

(ii) β particles (2 marks)

These are deflected toward the positive plate since the β particles are negatively charged.

(iii) γ rays (2 marks)

These remain undeviated since they have no charge.

(b) The equation below represents one nuclear fission reaction utilized in power plants.

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{141}_{56}Ba + ^{92}_{36}Kr + 3\,^{1}_{0}n + \gamma$$

With masses given as: $^{235}_{92}U = 3.902 \times 10^{-25} kg$

$^{1}_{0}n = 1.675 \times 10^{-27} kg$

$^{141}_{56}Ba = 2.289 \times 10^{-25} kg$

$^{92}_{36}Kr = 1.575 \times 10^{-25} kg$

(questions and answer on following page)

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Calculate

(i) The total mass of the particles (left hand side) and the products (right hand side) (4 marks)

Particles:

Mass of $^{235}_{92}U + ^{1}_{0}n = 3.902 \times 10^{-25}kg + 1.675 \times 10^{-27}kg$

$= 3.91875 \times 10^{-25}kg$

Products:

Mass of $^{141}_{56}Ba + ^{92}_{36}Kr + 3\,^{1}_{0}n = 2.289 \times 10^{-25} + 1.575 \times 10^{-25} +$

$3(1.675 \times 10^{-27})$

$= 3.91425 \times 10^{-25}kg$

(ii) The mass difference

$$\Delta m = 3.91875 \times 10^{-25} - 3.91425 \times 10^{-25}$$

$$= 4.5 \times 10^{-28}kg$$

(iii) The energy released in the nuclear reaction given that the speed of light is $c = 3.0 \times 10^{8}ms^{-1}$ (3 marks)

$$\Delta E = mc^2$$

$$= (4.5 \times 10^{-28})(3.0 \times 10^{8})^2$$

$$= 4.05 \times 10^{-11}J$$

Total 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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